

# Volkan Çiçek

Senior AI Engineer & Data Scientist

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## SUMMARY

Senior AI Engineer and Data Scientist with a Mathematics background and 5+ years at Logo Yazılım building production AI systems end to end. I work at both ends of the stack: designing and shipping LLM products (text-to-SQL, RAG, multi-agent systems) and owning the infrastructure they run on (Airflow/ETL, self-hosted GPU inference, MLOps). My text-to-SQL service runs in production across five ERP and HR products; the local LLM platform I built on NVIDIA DGX Spark serves 100+ users across the group; and through agent-driven engineering I delivered 80–90% of a legacy UI migration that consultancies had quoted at 5 to 12 months in under a month. MSc in Artificial Intelligence in progress.

## EXPERIENCE

### Senior Data Scientist

Logo Yazılım

Sep 2025 – Present

Gebze, Kocaeli (Hybrid)

- Own **PSA (Personalized SQL Agent)**, a production multi-tenant text-to-SQL service that turns natural language into SQL, NoSQL and function calls. Live across **five Logo ERP and HR products**: Tiger, Netsis, j-Platform, j-HR and Logo Lite. Built on FastAPI with async PaaS integrations, automatic query correction, retrieval and prompt-token optimisation, per-tenant isolation, rate limiting and credit metering; hardened through load, concurrency and production failure analysis.
- Designed, deployed and operate the **group-wide local LLM inference platform** on an NVIDIA DGX Spark (GB10, 128 GB unified memory), serving **100+ users**: vLLM running an NVFP4-quantised coding model behind a **LiteLLM** gateway, plus Ollama and Open WebUI for team chat. Authenticated, per-user metered access for Claude Code, Continue and VS Code Copilot, with Prometheus/Grafana monitoring and automated alerting. Benchmarked at ~ **1,100 tok/s across 32 parallel users (+70% after tuning)**; runs alongside the company's paid Copilot licences and keeps source code inside the network.
- Proved out **agent-driven engineering** across the group with three deliveries:
  - (1) migrated the UI of a UTM network-security appliance (Berqnet, Logo's cyber-security company) off a **331k-line, 2,500-file legacy Java/Vaadin** codebase onto a modern Nuxt/Vue frontend, reaching **80–90% ported in 2–3 weeks** against external consultancy bids of 5 to 12 months, then handed to the owning team to finish;
  - (2) built an **Agentic Development Platform PoC** to test whether the analyst team's agent-generated specs are code-ready: a multi-agent pipeline (plan / backend / frontend / integration / test / review) that produced a running Angular 17 + .NET 8 service against a live SOAP endpoint, with an auditable trace of every agent's input, output and open question as the actual deliverable;
  - (3) built the calculation core of a deterministic payroll engine for Turkish SGK/GİB legislation (.NET/C#, 22-step flow, 13 payroll types, effective-dated legislation parameters, every calculation verified to the kuruş with an xUnit suite) **in one week**, scoped to the agreed rule set rather than a finished product, on a problem where three divergent in-house applications and the requirements analysis itself had failed to converge.
- Built the **evaluation harness** that gates changes to PSA: a hand-curated per-product regression suite (Tiger, Netsis, j-Platform, j-HR, Logo Lite), an in-product **feedback loop** that grades answers against real user queries, and embedding A/B tests, so retrieval and prompt changes ship on measured accuracy, not intuition. Complemented by load, concurrency and penetration testing.
- Spearheaded a **centralised AI service architecture** for anomaly detection and forecasting, enabling cross-product integration and removing redundant per-product model deployments.
- Architected a **multi-agent PoC** using LLM orchestration to autonomously generate and validate dynamic UI components for security dashboards.
- Internal reference point for AI across the company**: brief teams on where the field is moving, run knowledge transfer on agentic development tooling (Claude Code, Cursor, Copilot), and work with the AI Transformation Lead on adoption.
- Mentor junior data scientists, interns and part-time staff**; set clean-code standards, own the team's AIOps and Generative AI roadmap, and turn business problems into deliverable projects with product and engineering counterparts.

### Data Scientist

Logo Yazılım

Jan 2022 – Sep 2025

Kocaeli, Türkiye

- Core contributor to **Mariana**, a TÜBİTAK TEYDEB-supported **AIOps** initiative: anomaly detection and forecasting over time-series telemetry from a **1,600-machine** estate, validated with hypothesis testing and RMSE/MAE monitoring to cut downtime.
- Engineered **ETL pipelines and Airflow orchestration** across two deployments for five cloud products (Logo Bulut ERP, Logo Bulut İK, Logo İşbaşı, Logo Lite, Logo Pera), powering multi-tenant licence, usage, sales and churn reporting sourced from REST APIs, PostgreSQL and web scrapers (Requests, BeautifulSoup, XPath).
- Led **Generative AI** delivery: intelligent chatbots, the first text-to-SQL engine, and Retrieval-Augmented Generation (RAG) systems for internal and customer-facing products.

- Built **AI code assistants and LLM-driven test-scenario generation** to shorten development cycles and raise software quality.
- Delivered **computer vision** in production: multi-model **OCR** pipelines for document digitisation integrated with PaaS services (Logo Flow, Logo HR), and a deep-learning initiative analysing real-time video feeds to detect workplace-safety risks before incidents occur.
- Built a desktop **data-labelling application** (MongoDB + CSV pipelines) for non-technical teams, and drove **customer segmentation** (RFM) projects from ideation to production monitoring.

#### Junior Data Scientist

Logo Yazılım

Jul 2021 – Dec 2021

Kocaeli, Türkiye

- Optimised an **OCR** pipeline for Turkish character recognition (Python, OpenCV, Pillow), improving accuracy and throughput.
- Built a **FastAPI** service for the Polaris project, migrating Atlassian documentation into Logo DYS and using **BERT/T5** transformers to auto-generate a Turkish Q&A dataset that backed an Azure Knowledge Base chatbot.

#### Freelance Web Developer

Self-employed

Jul 2015 – Jun 2021

Türkiye

- Custom websites and e-commerce platforms (WordPress, OpenCart) for small businesses and student organisations, owned end to end from client requirements through to hosting.

### EDUCATION

#### MSc, Artificial Intelligence (thesis, English)

Gebze Technical University

Sep 2025 – Jun 2027 (expected)

Gebze, Türkiye

#### BSc, Mathematics

Gebze Technical University

Sep 2018 – Jun 2022

Gebze, Türkiye

Thesis: *Principal Component Analysis* (advisor Prof. Dr. Nuri Çelik).

### TECHNICAL SKILLS

**Generative AI & Agents:** LLM application design, multi-agent orchestration, RAG, text-to-SQL, function/tool calling, MCP, prompt engineering, evaluation and guardrails, embeddings, LangChain, LangGraph, LlamaIndex, HuggingFace Transformers

**Models & Platforms:** Anthropic Claude, OpenAI, Azure OpenAI, Google Gemini, open-weight LLMs and vision-language models served locally

**Machine Learning:** Deep learning, time-series forecasting, anomaly detection, NLP, computer vision (OCR, VLMs), exploratory data analysis, customer segmentation (RFM), A/B testing, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, OpenCV

**Languages & Frameworks:** Python, SQL, C#/.NET, FastAPI, JavaScript/Vue/Nuxt, Bash, Git

**Data & Infrastructure:** Apache Airflow, ETL/ELT, PostgreSQL, MongoDB, DuckDB, vector databases (pgvector, Milvus), REST API design, microservices, multi-tenant architecture

**MLOps & Platform:** Docker, Kubernetes, CI/CD (Jenkins), MLflow, Prometheus + Grafana, vLLM, LiteLLM, Ollama, inference cost and token optimisation, Azure ML Studio, Proxmox virtualisation and self-hosted Linux infrastructure

**Ways of working:** Technical leadership, mentoring, Agile/Scrum, clean code and code review, technical writing (EN/TR)

**Languages:** Turkish (native), English (full professional proficiency)

### WRITING & OPEN SOURCE

- Technical blog at [volkancicek.com](https://volkancicek.com), where I publish long-form analysis of agentic engineering, LLM inference infrastructure and legacy-code modernisation (e.g. “Where Does an Agent Live?”, “What Can We Do With an NVIDIA DGX Spark?”).
- `litellm-supply-chain-audit`: an open-source CLI that statically detects the compromised LiteLLM PyPI releases from the March 2026 supply-chain attack, without importing the package.
- `VoiceDub Studio`, a self-built AI dubbing and voice-cloning product (TypeScript): designed, built and deployed solo, from product concept through to a live site.
- Upstream contributions to the tools I run in production. Diagnosed and reported a memory leak in `doctr` under FastAPI, an arm64 image shipping an amd64 binary in `nvidia_gpu_exporter`, and a version-bump gap in the GB10 vLLM build; merged pull requests to `daikin_onecta` and `HostsFileEditor`.

### CERTIFICATIONS & ADDITIONAL

- **Machine Learning Winter School**, Koç University & İş Bankası · **Mathematics for Machine Learning**, Nesin Mathematics Village · **NVIDIA Fundamentals of Deep Learning** · **BT Akademi DevOps**, Model Deployment, ML Model Validation, Container Architecture & Kubernetes
- **Chairman of the Board**, GTU Mathematics Club · **Financial Director**, IEEE GTU Student Branch · **Organisation Chief**, 9th Science & Technology Days (100-person team, 30 sponsors, 21 speakers)